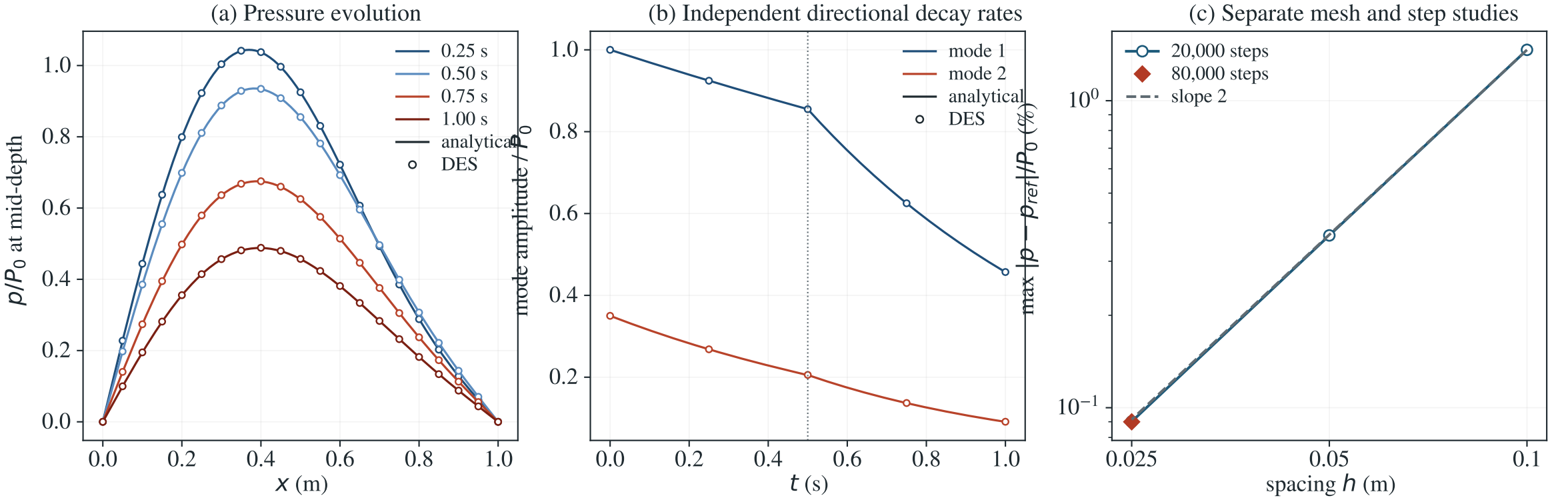

(a) Pressure evolution
0.25 s
0.50 s
0.75 s
1.00 s
analytical
DES
p/P_0 at mid-depth
x (m)
(b) Independent directional decay rates
mode 1
mode 2
analytical
DES
mode amplitude / P_0
t (s)
(c) Separate mesh and step studies
20,000 steps
80,000 steps
slope 2
max $|p - p_{ref}|/P_0$ (%)
spacing h (m)